MATH6183 – Regression Coursework

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Module: MATH6183 - Data Mining and Analytics

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1 Solutions

1.1 Problem 1

Consider the following sample:

$$(x_1, y_1) = (5, 5), (x_2, y_2) = (2, -5), (x_3, y_3) = (8, 12), (x_4, y_4) = (4, 1)$$

a. Suppose that you predict y based on x with a linear model $y = a + bx$ so that the mean squared error of the model is minimised. What are a and b? What is the $RMSE$ and R^2 of this model?

The simple linear regression model is given by $y = a + bx$, where a is the intercept and b is the slope. From provided data, $\bar{x}$ and $\bar{y}$ are calculated:

$$\bar{x} = \frac{5 + 2 + 8 + 4}{4} = 4.75$$

$$\bar{y} = \frac{5 - 5 + 12 + 1}{4} = 3.25$$

First, a and b are calculated using these equations:

$$\begin{aligned} b &= \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \\ &= \frac{(5-4.75)(5-3.25) + (8-4.75)(12-3.25) + (2-4.75)(-5-3.25) + (4-4.75)(1-3.25)}{(5-4.75)^2 + (8-4.75)^2 + (2-4.75)^2 + (4-4.75)^2} \\ &= 2.84 \end{aligned}$$

$$\begin{aligned} a &= \bar{y} - b\bar{x} \\ &= 3.25 - 2.84 \times 4.75 \\ &= -10.24 \end{aligned}$$

Replace $a = -10.24$ and $b = 2.84$ to $y = a + bx$ to obtain the estimated regression equation:

$$y = -10.24 + 2.84x \tag{1}$$

Next, substituting each x_i into (1), the predicted values of $\hat{y}$ are calculated as follows:

$$\begin{aligned} x_1 = 5 \quad &\Rightarrow \quad \hat{y_1} = 3.96 \\ x_2 = 2 \quad &\Rightarrow \quad \hat{y_2} = -4.56 \\ x_3 = 8 \quad &\Rightarrow \quad \hat{y_3} = 12.48 \\ x_4 = 4 \quad &\Rightarrow \quad \hat{y_4} = 1.12 \end{aligned}$$

Next, calculating R^2 and $RMSE$:

$$SSE = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = (5-3.96)^2 + (-5+4.56)^2 + (12-12.48)^2 + (1-1.12)^2 = 1.52$$

$$SST = \sum_{i=1}^{n}(y_i - \bar{y}_i)^2 = (5-3.25)^2 + (-5-3.25)^2 + (12-3.25)^2 + (1-3.25)^2 = 152.75$$

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{1.52}{152.75} \approx 0.9900490 \approx 99\%$$

$$RMSE = \sqrt{\frac{SSE}{n}} = \sqrt{\frac{1.52}{4}} \approx 0.61644$$

Results summary:

$$a = -10.24$$
$$b = 2.84$$
$$R^2 \approx 99\%$$
$$RMSE \approx 0.61644$$

b. Suppose that you predict instead with model $y = a + bz$, where z is a new feature that you defined to be $z = \sqrt{x}$. If you minimise the mean squared error of this model, what would be a and b? What is the $RMSE$ and R^2 of this model?

Because $z = \sqrt{x}$, the new linear regression model is $y = a + bz$

$z = \sqrt{x}$	$\sqrt{5}$	$\sqrt{2}$	$2\sqrt{2}$	2
y	5	-5	12	1

From the data provided, $\bar{z}$ and $\bar{y}$ are calculated:

$$\bar{z} = \frac{\sqrt{5} + \sqrt{2} + 2\sqrt{2} + 2}{4} = 2.12$$

$$\bar{y} = \frac{5 - 5 + 12 + 1}{4} = 3.25$$

Start with a and b

$$b = \frac{\sum_{i=1}^{n}(z_i - \bar{z})(y_i - \bar{y})}{\sum_{i=1}^{n}(z_i - \bar{z})^2}$$

$$= \frac{(\sqrt{5} - 2.12)(5 - 3.25) + (\sqrt{2} - 2.12)(-5 - 3.25) + (2\sqrt{2} - 2.12)(12 - 3.25) + (2 - 2.12)(1 - 3.25)}{(\sqrt{5} - 2.12)^2 + (\sqrt{2} - 2.12)^2 + (2\sqrt{2} - 2.12)^2 + (2 - 2.12)^2}$$

$$= 12.156$$

$$a = \bar{y} - b\bar{z}$$

$$= 3.25 - 12.156 \times 2.12$$

$$= -22.5207$$

Replace $a = -22.5207$ and $b = 12.156$ to $y = a + bz$ to obtain the estimated regression equation:

$$y = -22.5207 + 12.156z \tag{2}$$

Next, substituting each z_i into (2), the predicted values of $\hat{y}$ are calculated as follows:

$$z_1 = \sqrt{5} \quad \Rightarrow \quad \hat{y_1} = 4.66$$
$$z_2 = \sqrt{2} \quad \Rightarrow \quad \hat{y_2} = -5.33$$
$$z_3 = 2\sqrt{2} \quad \Rightarrow \quad \hat{y_3} = 11.86$$
$$z_4 = 2 \quad \Rightarrow \quad \hat{y_4} = 1.79$$

Next, calculating R^2 and $RMSE$:

$$SSE = \sum_{i=1}^{n}(y_i - \hat{y_i})^2 = (5 - 4.66)^2 + (-5 + 5.33)^2 + (12 - 11.86)^2 + (1 - 1.79)^2 = 0.866$$

$$SST = \sum_{i=1}^{n}(y_i - \bar{y_i})^2 = (5 - 3.25)^2 + (-5 - 3.25)^2 + (12 - 3.25)^2 + (1 - 3.25)^2 = 152.75$$

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{0.869}{152.75} \approx 0.99431 \approx 99.4\%$$

$$RMSE = \sqrt{\frac{SSE}{n}} = \sqrt{\frac{0.866}{4}} \approx 0.466$$

Results summary:

$$a = -22.52$$
$$b = 12.156$$
$$R^2 \approx 99.4\%$$
$$RMSE \approx 0.466$$

c. Draw the data points and the two models above on a graph. (Be careful to use a logarithmic scale where appropriate!)

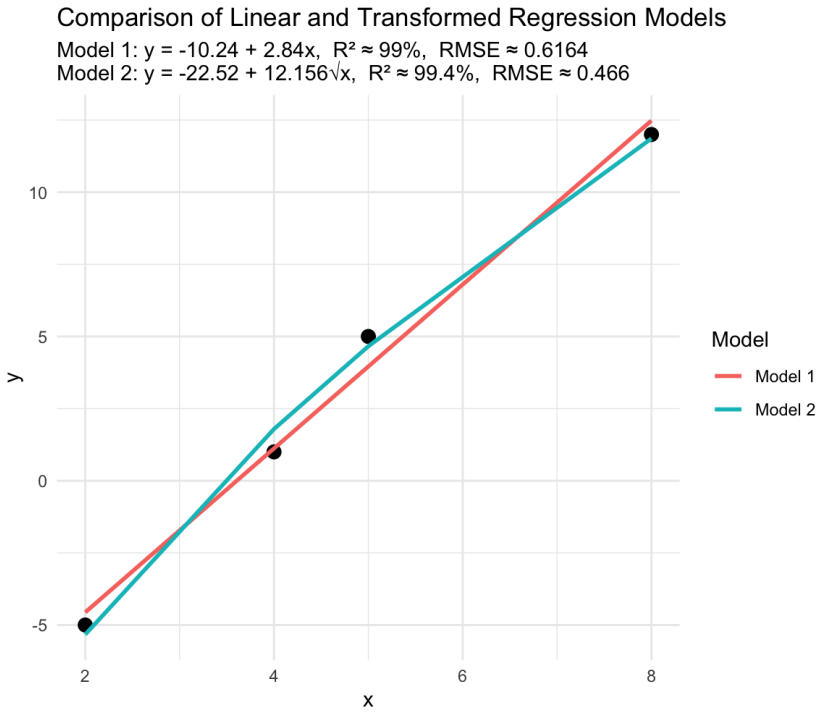

Figure 1: Comparison of Linear and Transformed Regression Models

Both models fit the data well with high R^2 values ($\approx$ 99%). However, the transformed model ($y = -22.52 + 12.156\sqrt{x}$) performs slightly better, achieving a smaller RMSE (0.466 vs 0.616).

```r
library(ggplot2)

# Data
data <- data.frame(
  x = c(5, 2, 8, 4),
  y = c(5, -5, 12, 1)
)

# Model 1: y = a + b*x
a1 <- -10.24
b1 <- 2.84
data$y_hat1 <- a1 + b1 * data$x

# Model 2: y = a + b*sqrt(x)
a2 <- -22.52
b2 <- 12.156
data$y_hat2 <- a2 + b2 * sqrt(data$x)

# Plot
ggplot(data, aes(x = x, y = y)) +
  geom_point(color = "black", size = 3) +
  geom_line(aes(y = y_hat1, color = "Model 1"), size = 1) +
  geom_line(aes(y = y_hat2, color = "Model 2"), size = 1) +
  labs(
    title = "Comparison of Linear and Transformed Regression Models",
    subtitle = "Model 1: y = -10.24 + 2.84x,  R²  99%,  RMSE  0.6164
    \nModel 2: y = -22.52 + 12.156x,  R²  99.4%,  RMSE  0.466",
    x = "x",
    y = "y",
    color = "Model"
  ) + theme_minimal()
  + theme(
    plot.title = element_text(face = "bold", hjust = 0.5, size = 13),
    plot.subtitle = element_text(hjust = 0.5, size = 11)
  )
```

1.2 Problem 2

For a certain type of cancer, the number y of malignant cells in a tissue sample is believed to grow exponentially with time t:

$$y = \alpha_0 e^{\alpha_1 t}$$

a. Is this model linear in the independent parameter t? If not, suggest a change of variables or transformation of y or t (or both!) that yields a linear relationship between the transformed variables.

The model $y = \alpha_0 e^{\alpha_1 t}$ is not linear in the independent t, this is an exponential model. To convert it to linear model, take the natural logarithm of both sides:

$$ln(y) = ln(\alpha_0) + \alpha_1 t \tag{3}$$

Let $z = ln(y)$, $a = ln(\alpha_0)$, $b = \alpha_1$, now the equation (3) takes the form $z = a + bt$
The model $z = a + bt$ is linear in the independent parameter t

b. Consider the observed data:

t (days)	1	2	3	4	5
y (cells)	16	27	45	74	122

Determine least squares estimates for the parameters α_0 and α_1 for this data. Draw the data points and the above model on a graph. What is the $RMSE$ and R^2 of this model?

Let $z = ln(y)$, $a = ln(\alpha_0)$, $b = \alpha_1$, the equation takes the form

$$z = a + bt$$

t	1	2	3	4	5
y	16	27	45	74	122
$z = ln(y)$	2.7726	3.2958	3.8067	4.3041	4.8040

From the data provided, $\bar{z}$, $\bar{y}$ and $\bar{t}$ are calculated:

$$\bar{z} = \frac{2.7726 + 3.2958 + 3.8067 + 4.3041 + 4.804}{5} = 3.79664$$

$$\bar{t} = \frac{1 + 2 + 3 + 4 + 5}{5} = 3$$

$$\bar{y} = \frac{16 + 27 + 45 + 74 + 122}{5} = 56.8$$

Calculating a and b:

$$b = \frac{\sum_{i=1}^{n}(t_i - \bar{t})(z_i - \bar{z})}{\sum_{i=1}^{n}(t_i - \bar{t})^2}$$

$$= \frac{(1-3)(2.7726 - 3.79664) + (2-3)(3.2958 - 3.79664) + (3-3)(3.8067 - 3.79664) + (4-3)(4.3041 - 3.79664) + (5-3)(4.8040 - 3.79664)}{(1-3)^2 + (2-3)^2 + (3-3)^2 + (4-3)^2 + (5-3)^2}$$

$$= 0.5071$$

$$\begin{aligned} a &= \bar{z} - b\bar{t} \\ &= 3.79664 - 0.5071 \times 3 \\ &= 2.2753099 \\ ln(\alpha_0) &= 2.2753099 \\ \rightarrow \alpha_0 &= e^{2.2753099} \approx 9.7309 \end{aligned}$$

Replace $a = 2.2753$ and $b = 0.5071$ to $z = a + bt$ to obtain the estimated regression equation:

$$z = 2.2753 + 0.5071t \tag{4}$$

Next, substituting each z_i into (4), the predicted values of $\hat{z}$ are calculated as follows:

$$\begin{aligned}
t_1 = 1 \quad &\Rightarrow \quad \hat{z_1} = 2.7824 \Rightarrow \quad \hat{y_1} = e^{2.7824} = 16.1579 \\
t_2 = 2 \quad &\Rightarrow \quad \hat{z_2} = 3.2895 \Rightarrow \quad \hat{y_2} = e^{3.2895} = 26.8297 \\
t_3 = 3 \quad &\Rightarrow \quad \hat{z_3} = 3.7966 \Rightarrow \quad \hat{y_3} = e^{3.7966} = 44.5498 \\
t_4 = 4 \Rightarrow \quad &\hat{z_4} = 4.3037 \Rightarrow \quad \hat{y_4} = e^{4.3037} = 73.9737 \\
t_5 = 5 \Rightarrow \quad &\hat{z_5} = 4.8108 \Rightarrow \quad \hat{y_5} = e^{4.8108} = 122.8310
\end{aligned}$$

Next, calculating R^2 and $RMSE$:

$$SSE = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = (16 - 16.1579)^2 + (27 - 26.8297)^2 + (45 - 44.5498)^2 + (74 - 73.9737)^2 + (122 - 122.8310)^2 = 0.9477$$

$$SST = \sum_{i=1}^{n}(y_i - \bar{y})^2 = (16 - 56.8)^2 + (27 - 56.8)^2 + (45 - 56.8)^2 + (74 - 56.8)^2 + (122 - 56.8)^2 = 7238.799$$

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{0.9477}{7238.799} \approx 0.99987 \text{ (or } 99.99\%)$$

$$RMSE = \sqrt{\frac{SSE}{n}} = \sqrt{\frac{0.9477}{5}} \approx 0.4354$$

Results summary:

$$\begin{aligned}
a &= 2.2753 \\
\alpha_0 &= 9.7309 \\
b = \alpha_1 &= 0.5071 \\
R^2 &\approx 99.99\% \\
RMSE &\approx 0.4354
\end{aligned}$$

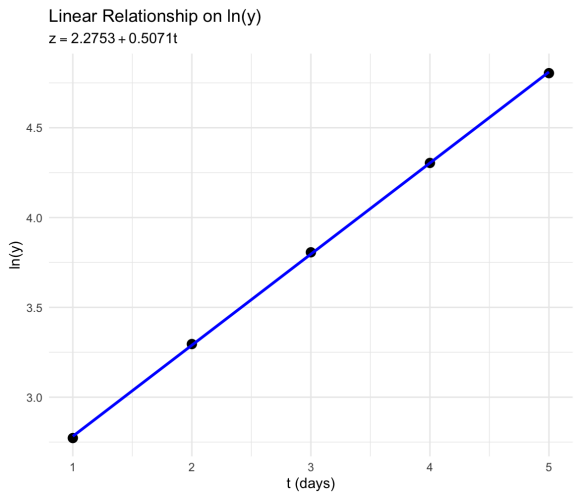

Figure 2: Linear Model on log(y)

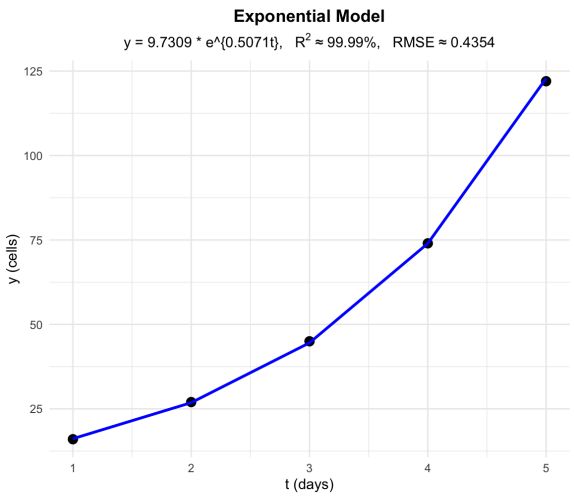

Figure 3: Exponential Model

Rcode for figure 2:

```
library(ggplot2)
data <- data.frame(
  t = c(1, 2, 3, 4, 5),
  y = c(16, 27, 45, 74, 122)
)
data$z <- log(data$y)
a <- 2.2753
b <- 0.5071
data$z_hat <- a + b * data$t
ggplot(data, aes(x = t, y = z)) +
  geom_point(size = 3, color = "black") +
  geom_line(aes(y = z_hat), color = "blue", size = 1) +
  labs(
    title = "Linear Relationship on ln(y)",
    subtitle = expression(z == 2.2753 + 0.5071*t),
    x = "t (days)",
    y = "ln(y)"
  ) +
  theme_minimal()
```

R code for figure 3:

```
library(ggplot2)

# Data
data <- data.frame(
  t = c(1, 2, 3, 4, 5),
  y = c(16, 27, 45, 74, 122)
)

# Model: y = a * exp(b * t)
a <- 9.7309
b <- 0.5071
```

```r
# Predict y hat
data$y_hat <- a * exp(b * data$t)

# Plot
ggplot(data, aes(x = t, y = y)) +
  geom_point(size = 3, color = "black") +
  geom_line(aes(y = y_hat), color = "blue", size = 1) +
  labs(
    title = "Exponential Model",
    subtitle = expression(paste("y = 9.7309 * e^{0.5071t},   R"^2, "  99.99%,   RMSE  0.4354")),
    x = "t (days)",
    y = "y (cells)"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5, size = 13),
    plot.subtitle = element_text(hjust = 0.5, size = 11)
  )
```

1.3 Problem 3

Consider $A \in \mathbb{R}^{m\times n}$ with $\operatorname{rank}(A) = n < m$ and $b \in \mathbb{R}^m$. Prove that x is a least squares solution of $Ax = b$ if and only if (y, x) is a solution of the larger system

$$\begin{pmatrix} I_{m\times m} & A \\ A^T & 0_{n\times n} \end{pmatrix} \begin{pmatrix} y \\ x \end{pmatrix} = \begin{pmatrix} b \\ 0 \end{pmatrix}.$$

From the above system equation

$$\begin{pmatrix} I_{m\times m} & A \\ A^T & 0_{n\times n} \end{pmatrix} \begin{pmatrix} y \\ x \end{pmatrix} = \begin{pmatrix} b \\ 0 \end{pmatrix},$$

Separating it into two equations:

$$\begin{cases} Iy + Ax = b & (1) \\ A^T y = 0 & (2) \end{cases}$$

From (1), infering that $y = b - Ax$, substituting this into the equation (2) gives

$$A^T(b - Ax) = 0.$$

Rearranging, it has the form

$$A^T(Ax - b) = 0.$$

The proof consists of the following two directions:

1. If x is a least squares solution, then $A^T(Ax - b) = 0$.
2. If $A^T(Ax - b) = 0$, then x is a least squares solution.

Start with direction (1), if x is a least squares solution of $Ax = b$, it minimises:

$$\begin{aligned} f(x) :&= \|Ax - b\|^2 = (Ax - b)^T(Ax - b) \\ &= A^T x^T Ax - A^T x^T b - b^T Ax + b^T b \\ &= A^T x^T Ax - 2x^T A^T b + b^T b \end{aligned}$$

Note: $A^T x^T b = b^T Ax = (x^T A^T b)^T$ because these expressions are scalars due to the given matrix dimensions (the transpose of a scalar is itself)

Let $Q = A^T A$ and $c = A^T b$, $f(x)$ becomes:

$$f(x) := Qxx^T - 2c^T x + b^T b$$

$$\begin{aligned} \nabla f(x) &= (Q + Q^T)x - 2c \\ &= 2Qx - 2c \\ &= 2A^T Ax - 2A^T b \end{aligned}$$

The minimum occurs when the derivative of $f(x)$ equals zero, that is,

$$\begin{aligned} \nabla f(x) &= 0 \\ 2A^T Ax - 2A^T b &= 0 \\ 2A^T Ax &= 2A^T b \\ A^T Ax &= A^T b \end{aligned}$$

Therefore, the least–squares solution x satisfies the **normal equation**

Regarding converse direction (2), suppose that $A^T(Ax - b) = 0$, x is proved to be a minimum point.

- Take any $u \in \mathbb{R}^n$.
- Let $y = u - x$ (the difference between any point u and the current solution x).

starting with $\|Au - b\|^2$, replacing $u = x + y$, achieving:

$$\begin{aligned} \|Au - b\|^2 &= \|A(x + y) - b\|^2 \\ f(u) &= \|Au - b\|^2 \\ &= (Au - b)^T(Au - b) \\ &= (A(x + y) - b)^T(A(x + y) - b) \\ &= (Ax + Ay - b)^T(Ax + Ay - b) \\ &= (Ax - b)^T(Ax - b) + 2y^T A^T(Ax - b) + (Ay)^T(Ay) \\ &= \|Ax - b\|^2 + 2y^T A^T(Ax - b) + \|Ay\|^2 \end{aligned}$$

Since $A^T(Ax - b) = 0$, giving

$$\|Au - b\|^2 = \|Ax - b\|^2 + \|Ay\|^2 \geq \|Ax - b\|^2$$

With any u values, $\|Au - b\|^2$ is always greater than or equal to $\|Ax - b\|^2 \Rightarrow x$ is the a least squares solution.

1.4 Problem 4

Consider the following sample:

$$(x_{11}, x_{12}, y_1) = (1, 1, 1), \quad (x_{21}, x_{22}, y_2) = (3, 0, 2), \quad (x_{31}, x_{32}, y_3) = (4, -1, 2).$$

a. If we fit a multiple linear regression model that uses both input variables x_1 and x_2 to this data, what would be the R^2? Explain your answer. (You can answer this question without calculating the model!)

With 3 given data points, matrix is defined as:

$$X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 3 & 0 \\ 1 & 4 & -1 \end{bmatrix}, y = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix},$$

In this case, the matrix is square matrix, computing the determinant of X:

- If $\det(A) \neq 0 \Rightarrow$ the equation has a unique solution.
- If $\det(A) = 0 \Rightarrow$ the equation has no solution or infinitely many solutions.

$$\begin{aligned} \det(X) &= 1 \begin{vmatrix} 3 & 0 \\ 4 & -1 \end{vmatrix} - 1 \begin{vmatrix} 1 & 0 \\ 1 & -1 \end{vmatrix} + 1 \begin{vmatrix} 1 & 3 \\ 1 & 4 \end{vmatrix} \\ &= 1(3 \cdot (-1) - 0 \cdot 4) - 1(1 \cdot (-1) - 0 \cdot 1) + 1(1 \cdot 4 - 3 \cdot 1) \\ &= (-3) - (-1) + (4 - 3) = -3 + 1 + 1 = -1 \neq 0 \end{aligned}$$

$\rightarrow$ X is full rank, Rank(X) = 3

Because $\det(X) \neq 0$, then X is invertible, meaning the regression model has a unique solution and $\boldsymbol{R^2 = 1}$ in the square case.

b. Assume another data point is added to your sample: $(x_{41}, x_{42}, y_4) = (0, 5, -1)$. Find the coefficients of the linear model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

that minimises the sum of squares error using normal equations. Justify that this solution is unique.

The coefficients of this model are calculated using normal equations:

$$\beta = (X^T X)^{-1} X^T y$$

With given data points,

$$X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 3 & 0 \\ 1 & 4 & -1 \\ 1 & 0 & 5 \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 2 \\ 2 \\ -1 \end{bmatrix}$$

where the first column of X corresponds to the intercept term.
To calculate normal equations, preparing the following matrices:

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 3 & 4 & 0 \\ 1 & 0 & -1 & 5 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 3 & 4 & 0 \\ 1 & 0 & -1 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 3 & 0 \\ 1 & 4 & -1 \\ 1 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 4 & 8 & 5 \\ 8 & 26 & -3 \\ 5 & -3 & 27 \end{bmatrix}$$

$$(X^T X)^{-1} = \begin{bmatrix} 4.5 & -1.5 & -1 \\ -1.5 & 0.539 & 0.338 \\ -1 & 0.338 & 0.260 \end{bmatrix}$$

$$X^T y \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 3 & 4 & 0 \\ 1 & 0 & -1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ 15 \\ -6 \end{bmatrix}$$

The normal equation is computed as follows:

$$\beta = (X^T X)^{-1} X^T y = \begin{bmatrix} 4.5 & -1.5 & -1 \\ -1.5 & 0.539 & 0.338 \\ -1 & 0.338 & 0.260 \end{bmatrix} \begin{bmatrix} 4 \\ 15 \\ -6 \end{bmatrix} = \begin{bmatrix} 1.5 \\ 0.057 \\ -0.49 \end{bmatrix}$$

The coefficients of the linear model:

$$\beta_0 = 1.5$$
$$\beta_1 = 0.057$$
$$\beta_2 = -0.49$$

1.5 Problem 5

The file `cheese_v_eng_phd.csv` compares the yearly per capita consumption of mozzarella cheese and the number of civil engineering doctorates awarded in the United States for 2000 through 2009.

a. Calculate the least squares linear regression line predicting the number of civil engineering doctorates y awarded as a function of the per capita mozzarella cheese consumption x.

- Report the slope and intercept of the regression line.
- Report the RMSE and R^2 for this linear model.
- Plot the regression line and data points on the same axis.

Let

$$\begin{cases} x: & \text{Yearly per capita consumption of mozzarella cheese,} \\ y: & \text{Number of civil engineering doctorates awarded in the United States.} \end{cases}$$

x	9.3	9.7	9.7	9.7	9.9	10.2	10.5	11	10.6
y	480	501	540	552	547	622	701	712	708

Linear model: $y = a + bx$ From provided data, $\bar{x}$ and $\bar{y}$ are calculated:

$$\bar{x} = \frac{9.3 + 9.7 + 9.7 + 9.7 + 9.9 + 10.2 + 10.5 + 11 + 10.6}{9} = 10.06667 \approx 10.07$$

$$\bar{y} = \frac{480 + 501 + 540 + 552 + 547 + 622 + 701 + 712 + 708}{9} = 595.88889 \approx 595.89$$

First, a and b are calculated using these calculations and equations:

$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$
−0.77	−115.89	88.85	0.59
−0.37	−94.89	34.79	0.13
−0.37	−55.89	20.49	0.13
−0.37	−43.89	16.09	0.13
−0.17	−48.89	8.15	0.03
0.13	26.11	3.48	0.02
0.43	105.11	45.55	0.19
0.93	116.11	108.37	0.87
0.53	112.11	59.79	0.28
		385.57	2.38

$$b = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{385.57}{2.38} \approx 162.00$$

$$a = \bar{y} - b\bar{x} = 595.89 - 162.00 \times 10.07 = -1035.45$$

Replace $a = -1035.45$ and $b = 162$ to $y = a + bx$ to obtain the estimated regression equation:

$$y = -1035.45 + 162x \tag{5}$$

Next, substituting each x_i into (5), the predicted values of $\hat{y}_i$ and the observed values are calculated as follows:

x_i	$\hat{y}_i$	$y_i - \hat{y}_i$	$y_i - \bar{y}_i$	$(y_i - \hat{y}_i)^2$	$(y_i - \bar{y}_i)^2$
9.3	471.15	8.85	−115.89	78.32	13,430.23
9.7	535.95	−34.95	−94.89	1,221.50	9,003.90
9.7	535.95	4.05	−55.89	16.40	3,123.57
9.7	535.95	16.05	−43.89	257.60	1,926.23
9.9	568.35	−21.35	−48.89	455.82	2,390.12
10.2	616.95	5.05	26.11	25.50	681.79
10.5	665.55	35.45	105.11	1,256.70	11,048.35
11	746.55	−34.55	116.11	1,193.70	13,481.79
10.6	681.75	26.25	112.11	689.06	12,568.90
				5,194.62	67,654.89

Next, calculating R^2 and $RMSE$:

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = 5,194.62$$

$$SST = \sum_{i=1}^{n} (y_i - \bar{y}_i)^2 = 67,654.89$$

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{5,194.62}{67,654.89} \approx 0.92322 \approx 92.32\%$$

$$RMSE = \sqrt{\frac{SSE}{n}} = \sqrt{\frac{5,194.62}{9}} \approx 24.025$$

Results summary:

$$a = -1035.45$$
$$b = 162.00$$
$$R^2 \approx 92.32\%$$
$$RMSE \approx 24.025$$

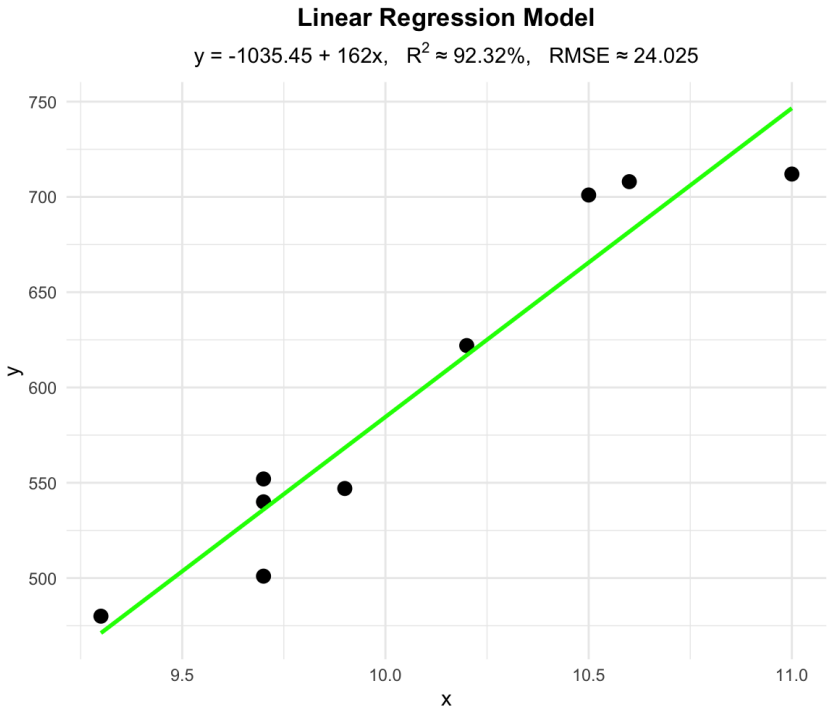

Figure 4: 5a-Linear Regression Model

```r
library(ggplot2)

# Data
data <- data.frame(
  x = c(9.3, 9.7, 9.7, 9.7, 9.9, 10.2, 10.5, 11, 10.6),
  y = c(480, 501, 540, 552, 547, 622, 701, 712, 708)
)

# Model: y = a + bx
a <- -1035.45
b <- 162.00
data$y_hat <- a + b * data$x

# Plot
ggplot(data, aes(x = x, y = y)) +
  geom_point(size = 3, color = "black") +
  geom_line(aes(y = y_hat), color = "green", size = 1) +
  labs(
    title = "Linear Regression Model",
    subtitle = expression(paste(
      "y = -1035.45 + 162x,   R"^2, "  92.32%,   RMSE  24.025"
    )),
    x = "x",
    y = "y"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5, size = 13),
```

```
  plot.subtitle = element_text(hjust = 0.5, size = 11)
)
```

b. Repeat the analysis of Part 5(a) using a **quadratic model**:

$$y = a + bx + cx^2.$$

x	9.3	9.7	9.7	9.7	9.9	10.2	10.5	11	10.6
x^2	86.49	94.09	94.09	94.09	98.01	104.04	110.25	121	112.36
y	480	501	540	552	547	622	701	712	708

The coefficients of this model are calculated using normal equations:

$$\beta = (X^T X)^{-1} X^T y$$

With given data points,

$$X = \begin{bmatrix} 1 & 9.3 & 86.49 \\ 1 & 9.7 & 94.09 \\ 1 & 9.7 & 94.09 \\ 1 & 9.7 & 94.09 \\ 1 & 9.9 & 98.01 \\ 1 & 10.2 & 104.04 \\ 1 & 10.5 & 110.25 \\ 1 & 11.0 & 121.00 \\ 1 & 10.6 & 112.36 \end{bmatrix}, \quad \beta = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \quad y = \begin{bmatrix} 480 \\ 501 \\ 540 \\ 552 \\ 547 \\ 622 \\ 701 \\ 712 \\ 708 \end{bmatrix}$$

where the first column of X corresponds to the intercept term
To form the normal equations, preparing the following matrices:

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 9.3 & 9.7 & 9.7 & 9.7 & 9.9 & 10.2 & 10.5 & 11.0 & 10.6 \\ 86.49 & 94.09 & 94.09 & 94.09 & 98.01 & 104.04 & 110.25 & 121.00 & 112.36 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 9.3 & 9.7 & 9.7 & 9.7 & 9.9 & 10.2 & 10.5 & 11.0 & 10.6 \\ 86.49 & 94.09 & 94.09 & 94.09 & 98.01 & 104.04 & 110.25 & 121.00 & 112.36 \end{bmatrix} \begin{bmatrix} 1 & 9.3 & 86.49 \\ 1 & 9.7 & 94.09 \\ 1 & 9.7 & 94.09 \\ 1 & 9.7 & 94.09 \\ 1 & 9.9 & 98.01 \\ 1 & 10.2 & 104.04 \\ 1 & 10.5 & 110.25 \\ 1 & 11.0 & 121.00 \\ 1 & 10.6 & 112.36 \end{bmatrix}$$

$$= \begin{bmatrix} 9.00 & 90.60 & 914.42 \\ 90.60 & 914.42 & 9253.52 \\ 914.42 & 9253.52 & 93890.42 \end{bmatrix}$$

$$(X^TX)^{-1} = \begin{bmatrix} 18867.50 & -3719.65 & 182.84 \\ -3719.65 & 733.73 & -36.09 \\ 182.84 & -36.09 & 1.78 \end{bmatrix}$$

$$X^Ty = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 9.3 & 9.7 & 9.7 & 9.7 & 9.9 & 10.2 & 10.5 & 11.0 & 10.6 \\ 86.49 & 94.09 & 94.09 & 94.09 & 98.01 & 104.04 & 110.25 & 121.00 & 112.36 \end{bmatrix} \begin{bmatrix} 480 \\ 501 \\ 540 \\ 552 \\ 547 \\ 622 \\ 701 \\ 712 \\ 708 \end{bmatrix} = \begin{bmatrix} 5363.0 \\ 54373.1 \\ 552713.0 \end{bmatrix}$$

The normal equation is computed as follows:

$$\beta = (X^TX)^{-1}X^Ty = \begin{bmatrix} 18867.50 & -3719.65 & 182.84 \\ -3719.65 & 733.73 & -36.09 \\ 182.84 & -36.09 & 1.78 \end{bmatrix} \begin{bmatrix} 5363.0 \\ 54373.1 \\ 552713.0 \end{bmatrix} = \begin{bmatrix} -3696.24 \\ 687.26 \\ -25.84 \end{bmatrix}$$

Results:

$$a = -3696.24$$
$$b = 687.26$$
$$c = -25.84$$

Achieving the model $y = -3696.24 + 687.26x - 25.84x^2$
To prepare the necessary metrics for the model, computing the following values:

x_i	$\hat{y}_i$	$y_i - \hat{y}_i$	$y_i - \bar{y}_i$	$(y_i - \hat{y}_i)^2$	$(y_i - \bar{y}_i)^2$
9.3	460.38	19.62	−115.89	385.09	13430.23
9.7	538.90	−37.90	−94.89	1436.14	9003.90
9.7	538.90	1.10	−55.89	1.22	3123.57
9.7	538.90	13.10	−43.89	171.70	1926.23
9.9	575.06	−28.06	−48.89	787.12	2390.12
10.2	625.42	−3.42	26.11	11.69	681.79
10.5	671.13	29.87	105.11	892.22	11048.35
11.0	736.98	−24.98	116.11	624.00	13481.79
10.6	685.33	22.67	112.11	513.77	12568.90
				4822.93	67654.89

Next, calculating R^2 and $RMSE$:

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = 4822.93$$

$$SST = \sum_{i=1}^{n} (y_i - \bar{y}_i)^2 = 67654.89$$

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{4822.93}{67654.89} \approx 0.92871 \approx 92.87\%$$

$$RMSE = \sqrt{\frac{SSE}{n}} = \sqrt{\frac{4822.93}{9}} \approx 23.15$$

Results summary:

$$a = -3696.24$$
$$b = 687.26$$
$$c = -25.84$$
$$R^2 \approx 92.87\%$$
$$RMSE \approx 23.15$$

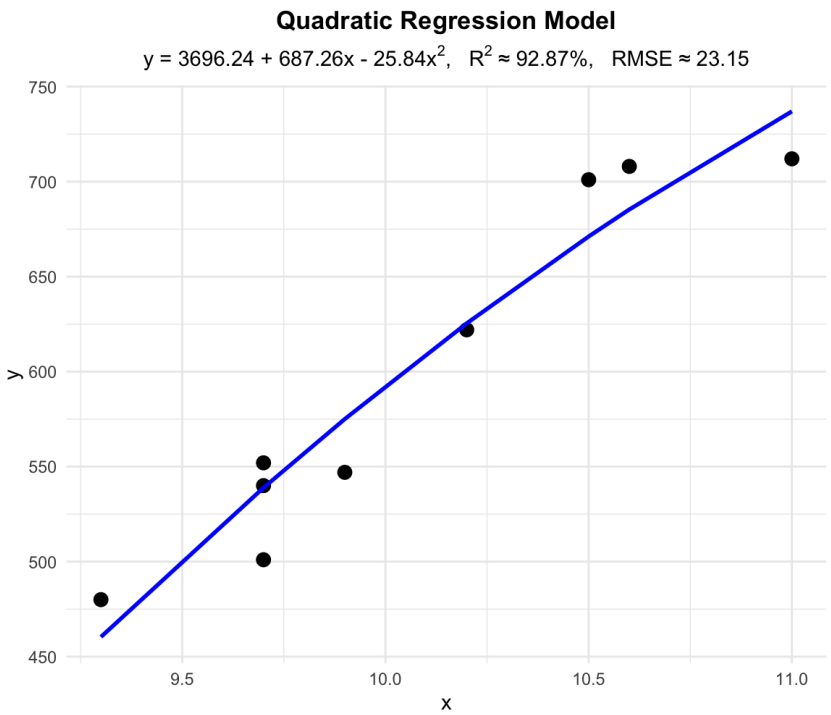

Figure 5: 5b-Quadratic Model

```r
library(ggplot2)

# Data
data <- data.frame(
  x = c(9.3, 9.7, 9.7, 9.7, 9.9, 10.2, 10.5, 11, 10.6),
  y = c(480, 501, 540, 552, 547, 622, 701, 712, 708)
)

# Coefficients from results summary
a <- -3696.24
b <- 687.26
c <- -25.84

# Predicted values (quadratic model)
data$y_hat <- a + b * data$x + c * data$x^2

# Plot
ggplot(data, aes(x = x, y = y)) +
  geom_point(size = 3, color = "black") +
  geom_line(aes(y = y_hat), color = "blue", size = 1) +
  labs(
    title = "Quadratic Regression Model",
    subtitle = expression(paste(
      "y = 3696.24 + 687.26x - 25.84x"^2,
      ",   R"^2, "  92.87%,   RMSE  23.15"
    )),
    x = "x",
    y = "y"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5, size = 13),
    plot.subtitle = element_text(hjust = 0.5, size = 11)
  )
```

c. Repeat the analysis of Parts 5(a) and 5(b) using a **cubic model**:

$$y = a + bx + cx^2 + dx^3.$$

x	9.3	9.7	9.7	9.7	9.9	10.2	10.5	11	10.6
x^2	86.49	94.09	94.09	94.09	98.01	104.04	110.25	121	112.36
x^3	804.36	912.67	912.67	912.67	970.30	1061.21	1157.63	1331	1191.016
y	480	501	540	552	547	622	701	712	708

The coefficients of this model are calculated using normal equations:

$$\beta = (X^T X)^{-1} X^T y$$

With given data points,

$$X = \begin{bmatrix} 1 & 9.3 & 86.49 & 804.36 \\ 1 & 9.7 & 94.09 & 912.67 \\ 1 & 9.7 & 94.09 & 912.67 \\ 1 & 9.7 & 94.09 & 912.67 \\ 1 & 9.9 & 98.01 & 970.30 \\ 1 & 10.2 & 104.04 & 1061.21 \\ 1 & 10.5 & 110.25 & 1157.63 \\ 1 & 11.0 & 121.00 & 1331.00 \\ 1 & 10.6 & 112.36 & 1191.02 \end{bmatrix} \quad \beta = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}, \quad y = \begin{bmatrix} 480 \\ 501 \\ 540 \\ 552 \\ 547 \\ 622 \\ 701 \\ 712 \\ 708 \end{bmatrix}$$

where the first column of X corresponds to the intercept term

To compute the normal equation to find a, b, c, d, preparing the matrices as follows:

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 9.3 & 9.7 & 9.7 & 9.7 & 9.9 & 10.2 & 10.5 & 11.0 & 10.6 \\ 86.49 & 94.09 & 94.09 & 94.09 & 98.01 & 104.04 & 110.25 & 121.00 & 112.36 \\ 804.36 & 912.67 & 912.67 & 912.67 & 970.30 & 1061.21 & 1157.63 & 1331.00 & 1191.02 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 9.3 & 9.7 & 9.7 & 9.7 & 9.9 & 10.2 & 10.5 & 11.0 & 10.6 \\ 86.49 & 94.09 & 94.09 & 94.09 & 98.01 & 104.04 & 110.25 & 121.00 & 112.36 \\ 804.36 & 912.67 & 912.67 & 912.67 & 970.30 & 1061.21 & 1157.63 & 1331.00 & 1191.02 \end{bmatrix} \begin{bmatrix} 1 & 9.3 & 86.49 & 804.36 \\ 1 & 9.7 & 94.09 & 912.67 \\ 1 & 9.7 & 94.09 & 912.67 \\ 1 & 9.7 & 94.09 & 912.67 \\ 1 & 9.9 & 98.01 & 970.30 \\ 1 & 10.2 & 104.04 & 1061.21 \\ 1 & 10.5 & 110.25 & 1157.63 \\ 1 & 11.0 & 121.00 & 1331.00 \\ 1 & 10.6 & 112.36 & 1191.02 \end{bmatrix}$$

$$= \begin{bmatrix} 9.000 & 90.600 & 914.420 & 9253.524 \\ 90.600 & 914.420 & 9253.524 & 93890.418 \\ 914.420 & 9253.524 & 93890.418 & 955197.844 \\ 9253.524 & 93890.418 & 955197.844 & 9743724.520 \end{bmatrix}$$

$$(X^T X)^{-1} = \begin{bmatrix} 6458649.483 & -1915106.815 & 188979.052 & -6205.786 \\ -1915106.815 & 568051.207 & -56072.558 & 1841.935 \\ 188979.052 & -56072.558 & 5536.747 & -181.936 \\ -6205.786 & 1841.935 & -181.936 & 5.980 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 9.3 & 9.7 & 9.7 & 9.7 & 9.9 & 10.2 & 10.5 & 11.0 & 10.6 \\ 86.49 & 94.09 & 94.09 & 94.09 & 98.01 & 104.04 & 110.25 & 121.00 & 112.36 \\ 804.36 & 912.67 & 912.67 & 912.67 & 970.30 & 1061.21 & 1157.63 & 1331.00 & 1191.02 \end{bmatrix} \begin{bmatrix} 480 \\ 501 \\ 540 \\ 552 \\ 547 \\ 622 \\ 701 \\ 712 \\ 708 \end{bmatrix} = \begin{bmatrix} 5363.0 \\ 54373.1 \\ 552713.0 \\ 5633210.8 \end{bmatrix}$$

The normal equation is computed:

$$\beta = (X^T X)^{-1} X^T y = \begin{bmatrix} 6458649.483 & -1915106.815 & 188979.052 & -6205.786 \\ -1915106.815 & 568051.207 & -56072.558 & 1841.935 \\ 188979.052 & -56072.558 & 5536.747 & -181.936 \\ -6205.786 & 1841.935 & -181.936 & 5.980 \end{bmatrix} \begin{bmatrix} 5363.0 \\ 54373.1 \\ 552713.0 \\ 5633210.8 \end{bmatrix} = \begin{bmatrix} 128935.43 \\ -38679.06 \\ 3862.54 \\ -127.81 \end{bmatrix}$$

Results:

$$a = 128935.43$$
$$b = -38679.06$$
$$c = 3862.54$$
$$d = -127.81$$

Achieving the model $y = 128935.4340 - 38679.06x + 3862.54x^2 - 127.81x^3$
To prepare the necessary metrics for the model, computing the following values:

x_i	$\hat{y}_i$	$y_i - \hat{y}_i$	$y_i - \bar{y}_i$	$(y_i - \hat{y}_i)^2$	$(y_i - \bar{y}_i)^2$
9.3	486.39	−6.39	−115.89	40.86	13430.23
9.7	526.20	−25.20	−94.89	635.27	9003.90
9.7	526.20	13.80	−55.89	190.32	3123.57
9.7	526.20	25.80	−43.89	665.41	1926.23
9.9	566.37	−19.37	−48.89	375.21	2390.12
10.2	634.69	−12.69	26.11	161.01	681.79
10.5	694.29	6.71	105.11	45.05	11048.35
11.0	718.00	−6.00	116.11	36.05	13481.79
10.6	708.64	−0.64	112.11	0.41	12568.90
				2149.58	67654.89

Next, calculating R^2 and $RMSE$:

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = 2149.58$$

$$SST = \sum_{i=1}^{n} (y_i - \bar{y}_i)^2 = 67654.89$$

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{2149.58}{67654.89} \approx 0.96823 \approx 96.82\%$$

$$RMSE = \sqrt{\frac{SSE}{n}} = \sqrt{\frac{2149.58}{9}} \approx 15.45$$

Results summary:

$$a = 128935.43$$
$$b = -38679.06$$
$$c = 3862.54$$
$$d = -127.81$$
$$R^2 \approx 96.82\%$$
$$RMSE \approx 15.45$$

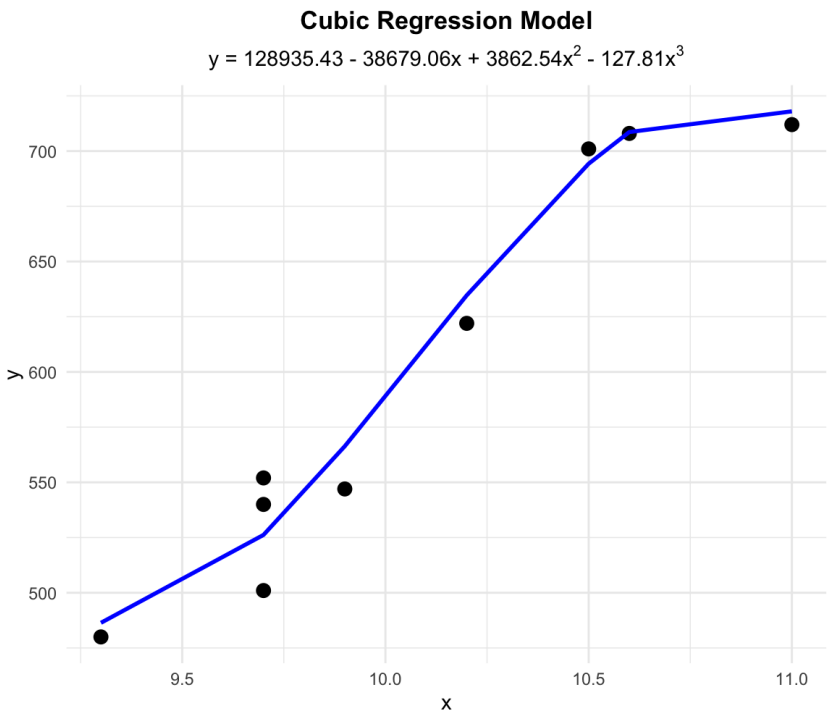

Figure 6: 5c-Cubic Model

```r
library(ggplot2)

# Data
data <- data.frame(
  x = c(9.3, 9.7, 9.7, 9.7, 9.9, 10.2, 10.5, 11, 10.6),
  y = c(480, 501, 540, 552, 547, 622, 701, 712, 708)
)

# Coefficients (from results)
a <- 128935.43
b <- -38679.06
c <- 3862.54
d <- -127.81

# Predicted values (cubic model)
data$y_hat <- a + b * data$x + c * data$x^2 + d * data$x^3

# Plot
ggplot(data, aes(x = x, y = y)) +
  geom_point(size = 3, color = "black") +
  geom_line(aes(y = y_hat), color = "blue", size = 1) +
  labs(
    title = "Cubic Regression Model",
    subtitle = expression(paste(
      "y = 128935.43 - 38679.06x + 3862.54x"^2, " - 127.81x"^3
    )),
    x = "x",
    y = "y"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5, size = 13),
    plot.subtitle = element_text(hjust = 0.5, size = 11)
  )
```

d. Which of the three models from Parts 5(a), 5(b), and 5(c) should we use to predict the number of PhDs awarded from unseen annual mozzarella consumption figures? Which do you think is most prone to overfitting?

Looking back 3 models that are built in 5a, 5b and 5c:

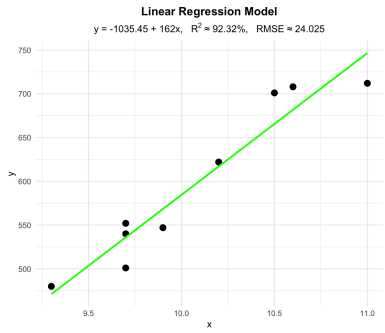

Figure 7: 5a - Linear Regression Model

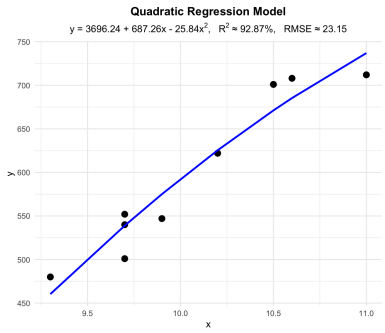

Figure 8: 5b - Quadratic Regression Model

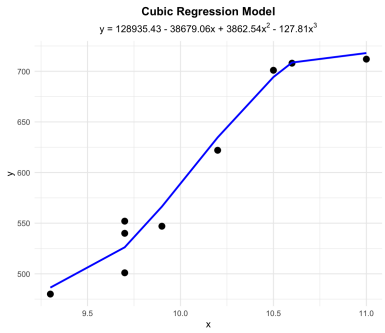

Figure 9: 5c - Cubic Regression Model

	Linear Regression	Quadratic Regression	Cubic Regression
R^2 (%)	92.32	92.87	96.82
$RMSE$	24.025	23.15	15.45

Across the three models, all show strong performance with $R^2 > 92\%$, meaning the independent variable explains over 92% of the variation in the dependent variable. The RMSE values are within the range (15, 25). Although the data points are not perfectly linear, they still exhibit a strong positive correlation overall.

These results suggest that the Quadratic Regression model can be used to predict because:

- The quadratic model effectively captures the nonlinear pattern without excessive complexity, providing a good balance between model fit and generalisation ability.
- The linear model is less flexible and in this case, it may not fully represent the relationship between variables. However, it is simple, interpretable, and computationally efficient.
- The cubic model is unnecessarily complex for this case.

Most likely prone to overfitting: Cubic Regression

Overfitting occurs when a model fits the training data very well (low RMSE in the training dataset) but performs poorly on unseen data (high RMSE in the test dataset), often because it captures random noise rather than true underlying patterns. More flexible models generally have a higher chance of overfitting.

Possible explanations for cubic regression are:

- It has the largest number of parameters (four compared with three for the quadratic and two for the linear model), making overfitting more likely.
- Very sensitive to outliers due to the x^3 term, which can cause large fluctuations in fitted values.
- In practice, there is no theoretical reason to expect the relationship between mozzarella consumption and the number of PhD degrees to follow a cubic form.

However, cubic regression offers higher flexibility, fitting highly curved relationships with multiple inflexion points.

To avoid overfitting in real applications, reasonable approaches include cross-validation, bias–variance trade-off analysis, feature selection, and other evaluation techniques used to assess model performance on new data objectively.

e. Do you think that the results of the previous parts indicate a meaningful relationship between these two statistics, i.e., does increased cheese consumption lead to more engineering doctorates?

These two variables are highly correlated. However, there is probably no meaningful relationship between cheese consumption and engineering doctorates. In this case, these variables may have a **spurious correlation**. Probable explanations are that:

- Correlation is not causation. There is insufficient evidence to conclude that these two variables have a cause–and–effect relationship.
- There might be a confounding variable that affects both variables.
- The data points come from small samples that do not represent the whole population, which can create a false relationship.
- Measurement errors could also produce a misleading association.

In reality, if cheese consumption truly caused an increase in engineering doctorates, governments would promote eating more cheese to achieve educational targets, which is not logically consistent.

Strategies to identify a true cause–and–effect relationship might involve conducting controlled experiments or identifying omitted variables that explain the observed association.

2 References

James, G., Witten, D., Hastie, T. and Tibshirani, R. (2021) *An Introduction to Statistical Learning.* New York, USA: Springer Science+Business Media, LLC, part of Springer Nature.

Buisson, F. (2021) *Behavioral Data Analysis with R and Python.* Sebastopol, CA: O'Reilly Media, Inc.